



datamining_assignment

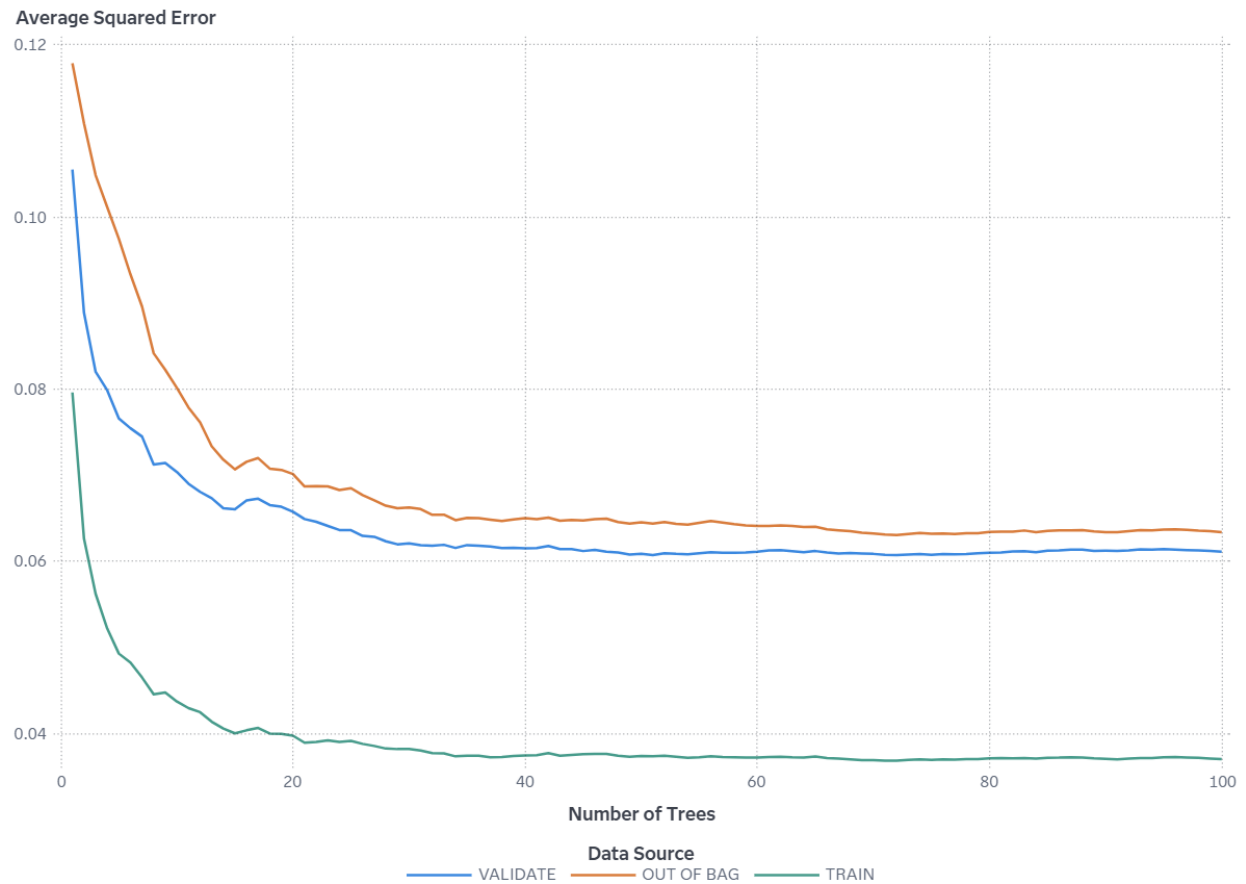
"Forest" Results

by: tp079964@mail.apu.edu.my

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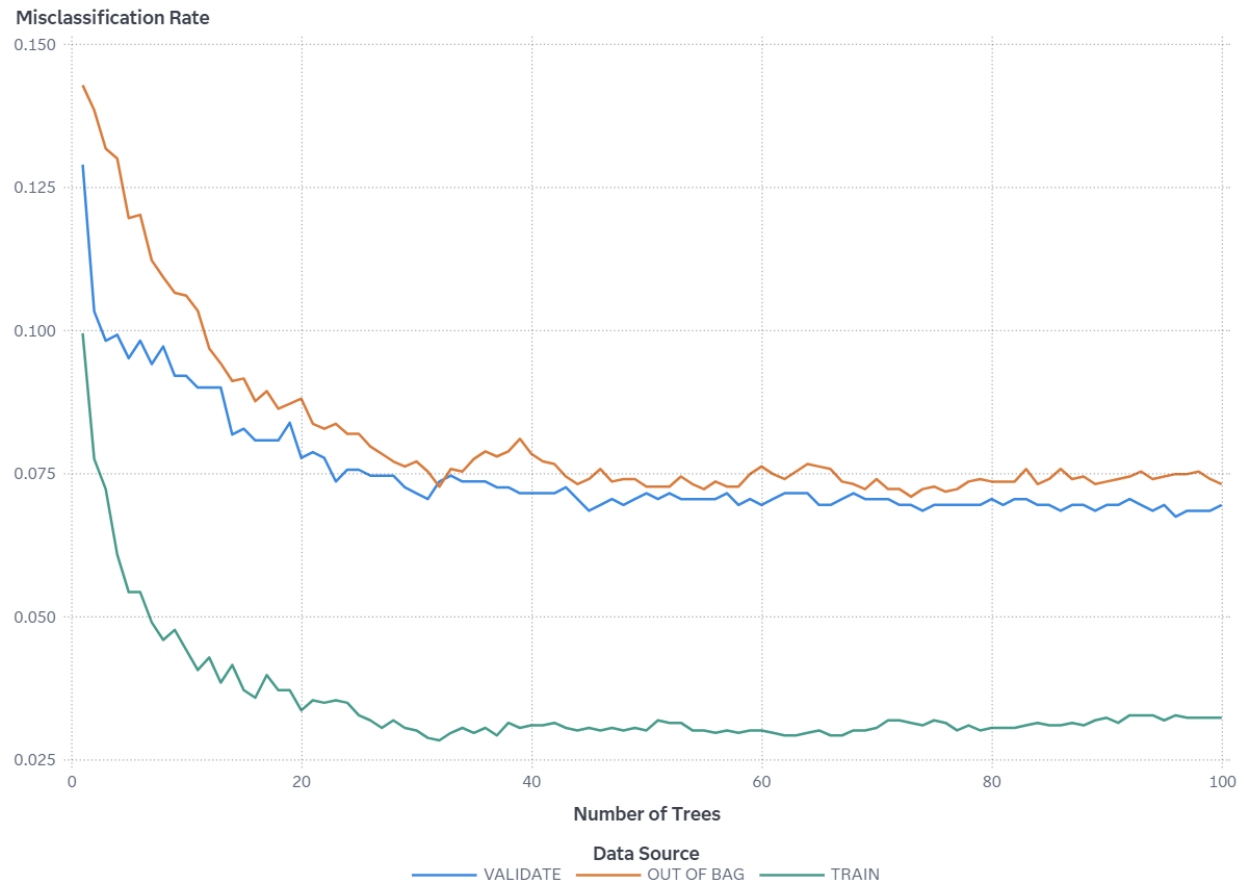
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Average Squared Error



This plot shows how the average squared error changes as the number of trees in the forest increases. The training error typically decreases as the number of trees increases, but the error for the VALIDATE partition gives you an indication of how well your model generalizes. For this model, the minimum error for the VALIDATE partition is 0.061 and occurs for 51 trees.

Misclassification Rate



This plot shows how the misclassification rate changes as the number of trees in the forest increases. The training error typically decreases as the number of trees increases, but the error for the VALIDATE partition gives you an indication of how well your model generalizes. For this model, the minimum error for the VALIDATE partition is 0.068 and occurs for 96 trees.

Variable Importance

Variable Label	Role	Variable Name	Training Importance
	INPUT	Total_Trans_Ct	161.2474
	INPUT	Total_Trans_Amt	123.7724
	INPUT	Total_Revolving_Bal	76.4430
	INPUT	Total_Ct_Chng_Q4_Q1	34.3428
	INPUT	Total_Relationship_Count	29.2332
	INPUT	Avg_Utilization_Ratio	27.4702
	INPUT	Total_Amt_Chng_Q4_Q1	25.9366
	INPUT	Contacts_Count_12_mon	19.6626
	INPUT	Months_Inactive_12_mon	16.1134
	INPUT	Customer_Age	12.7258
	INPUT	Avg_Open_To_Buy	10.7093
	INPUT	Credit_Limit	8.8243
	INPUT	Education_Level	8.4031
	INPUT	Months_on_book	8.1430
	INPUT	Dependent_count	6.9021
	INPUT	Income_Category	5.8444
	INPUT	Marital_Status	3.2964
	INPUT	Card_Category	2.1565
	INPUT	Gender	1.3241

Importance Standard Deviation	Relative Importance
78.4780	1
72.4968	0.7676
49.6597	0.4741
21.9636	0.2130
13.2312	0.1813
33.2783	0.1704
12.8737	0.1608
9.4514	0.1219
12.7437	0.0999
7.7651	0.0789
7.7634	0.0664
5.8564	0.0547
5.0302	0.0521
5.5882	0.0505
4.2941	0.0428
4.3855	0.0362
2.7318	0.0204
2.7196	0.0134
2.0154	0.0082

Score Inputs

Name	Role	Variable Level	Type
Avg_Open_To_Buy	INPUT	INTERVAL	N
Avg_Utilization_Ratio	INPUT	INTERVAL	N
Card_Category	INPUT	NOMINAL	C
Contacts_Count_12_mon	INPUT	NOMINAL	N
Credit_Limit	INPUT	INTERVAL	N
Customer_Age	INPUT	INTERVAL	N
Dependent_count	INPUT	NOMINAL	N
Education_Level	INPUT	NOMINAL	C
Gender	INPUT	BINARY	C
Income_Category	INPUT	NOMINAL	C
Marital_Status	INPUT	NOMINAL	C
Months_Inactive_12_mon	INPUT	NOMINAL	N
Months_on_book	INPUT	INTERVAL	N
Total_Amt_Chng_Q4_Q1	INPUT	INTERVAL	N
Total_Ct_Chng_Q4_Q1	INPUT	INTERVAL	N
Total_Relationship_Count	INPUT	NOMINAL	N
Total_Revolving_Bal	INPUT	INTERVAL	N
Total_Trans_Amt	INPUT	INTERVAL	N
Total_Trans_Ct	INPUT	INTERVAL	N

Variable Type	Variable Label	Variable Format	Variable Length
double			8
double			8

Score Outputs

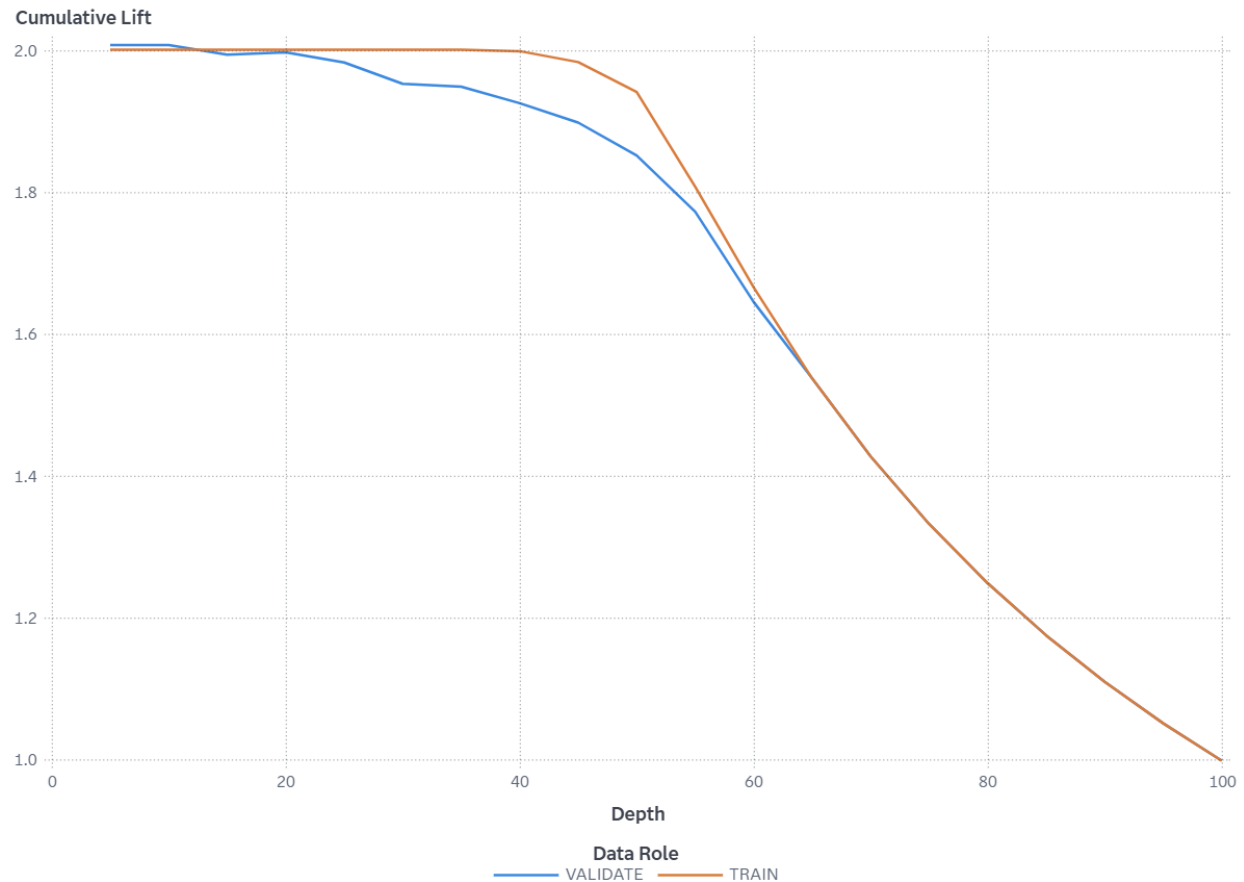
Name	Role	Type	Variable Type
EM_CLASSIFICATION	CLASSIFICATION	C	char
EM_EVENTPROBABILITY	PREDICT	N	double
EM_PROBABILITY	PREDICT	N	double
I_Attrition_Flag	CLASSIFICATION	C	char
P_Attrition_FlagAttrited_Custome	PREDICT	N	double
P_Attrition_FlagExisting_Custome	PREDICT	N	double
WARN	ASSESS	C	char

Variable Label	Variable Format	Variable Length	Creator
Predicted for Attrition_Flag		17	forest
Probability for Attrition_Flag=Attrited Customer		8	forest
Probability of Classification		8	forest
Into: Attrition_Flag		17	forest
Predicted: Attrition_Flag=Attrited Customer		8	forest
Predicted: Attrition_Flag=Existing Customer		8	forest
Warnings		4	forest

Function	Creator GUID
CLASSIFICATION	2afab3fe-bd57-4f14-

Function	Creator GUID
	a323-14ed8c04bb2 a
PREDICT	2afab3fe- bd57-4f14- a323-14ed8c04bb2 a
PREDICT	2afab3fe- bd57-4f14- a323-14ed8c04bb2 a
CLASSIFICATION	2afab3fe- bd57-4f14- a323-14ed8c04bb2 a
PREDICT	2afab3fe- bd57-4f14- a323-14ed8c04bb2 a
PREDICT	2afab3fe- bd57-4f14- a323-14ed8c04bb2 a
ASSESS	2afab3fe- bd57-4f14- a323-14ed8c04bb2 a

Cumulative Lift



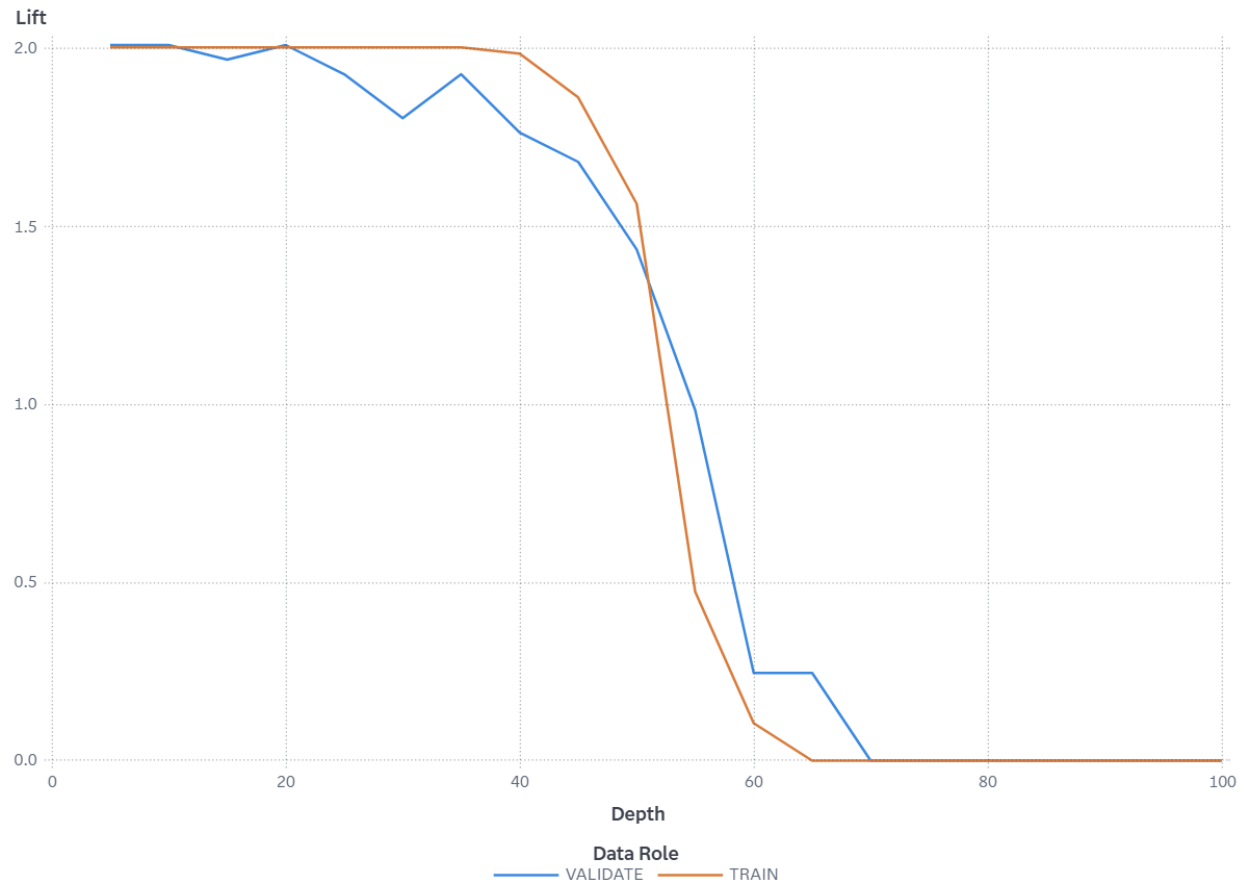
The VALIDATE partition has a Cumulative Lift of 2.01 in the 10% quantile (depth of 10) meaning there are 2.01 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TRAIN partition has a Cumulative Lift of 2 in the 10% quantile (depth of 10) meaning there are 2 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

Cumulative lift is calculated by sorting each partition in descending order by the predicted probability of the target event `P_Attrition_FlagAttrited_Customer`, which represents the predicted probability of the event "Attrited Customer" for the target `Attrition_Flag`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative lift for a

particular quantile is the ratio of the number of events across all quantiles up to and including the current quantile to the number of events that would be there at random, or equivalently, the ratio of the cumulative response percentage to the baseline response percentage. The cumulative lift at depth 10 includes the top 10% of the data, which is the first 2 quantiles, which would have 10% of the events at random. Thus, cumulative lift measures how much more likely it is to observe an event in the quantiles than by selecting observations at random.

Lift



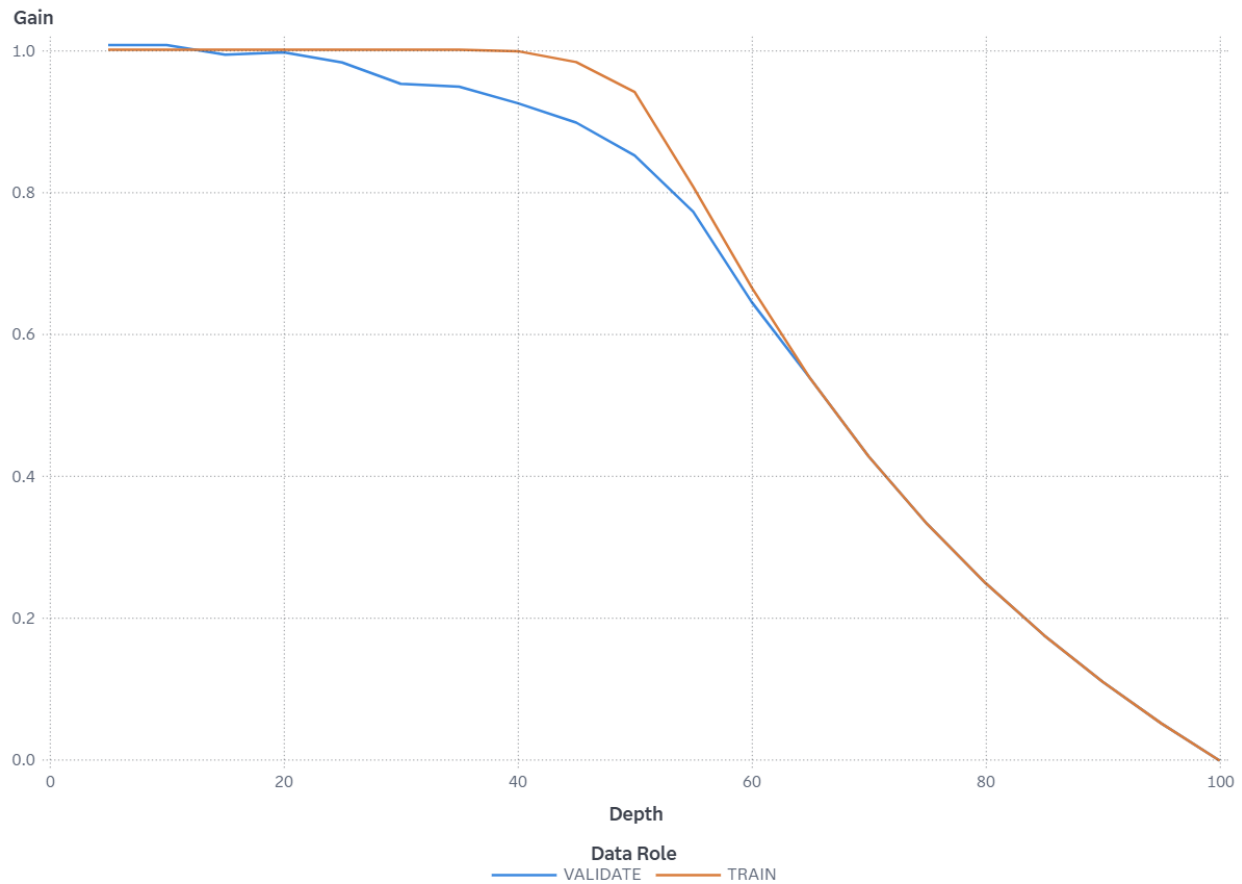
The VALIDATE partition has a Lift of 2.01 in the 5% quantile (depth of 5) meaning there are 2.01 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TRAIN partition has a Lift of 2 in the 5% quantile (depth of 5) meaning there are 2 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

Lift is calculated by sorting each partition in descending order by the predicted probability of the target event `P_Attrition_FlagAttrited_Custome`, which represents the predicted probability of the event "Attrited Customer" for the target `Attrition_Flag`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Lift is the ratio of the number of events in that quantile to the number of events that would be there at random, or equivalently, the ratio of the response percentage to the baseline response percentage. With 20

quantiles, it is expected that 5% of the events occur in each quantile. Thus, Lift measures how much more likely it is to observe an event in each quantile than by selecting observations at random.

Gain



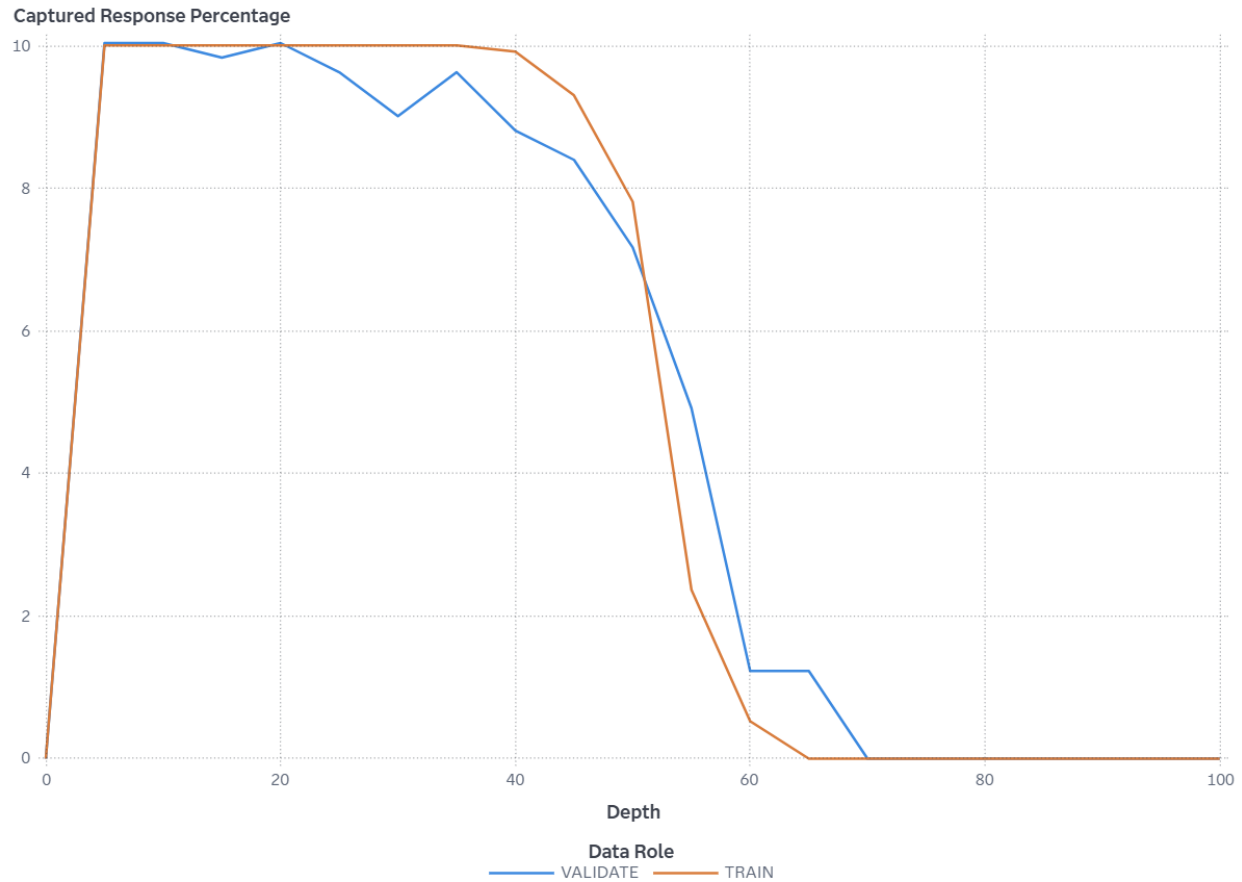
The VALIDATE partition has a Gain of 1 at the 10% quantile (depth of 10). Because this value is greater than 0, it is better to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 1.01.

The TRAIN partition has a Gain of 1 at the 10% quantile (depth of 10). Because this value is greater than 0, it is better to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 1.

Gain is calculated by sorting each partition in descending order by the predicted probability of the target event `P_Attrition_FlagAttrited_Customers`, which represents the predicted probability of the event "Attrited Customer" for the target `Attrition_Flag`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Gain is a cumulative measure for the quantiles up to and including the current one and is calculated as $(\text{number of events in the quantiles}) / (\text{number of events expected by random}) - 1$. With 20 quantiles, it is

expected that 5% of the events occur in each quantile. Note that the value of Gain is the same as the value of Cumulative Lift - 1. If the value of Gain is greater than 0, then your model is better at identifying events than using no model.

Captured Response Percentage

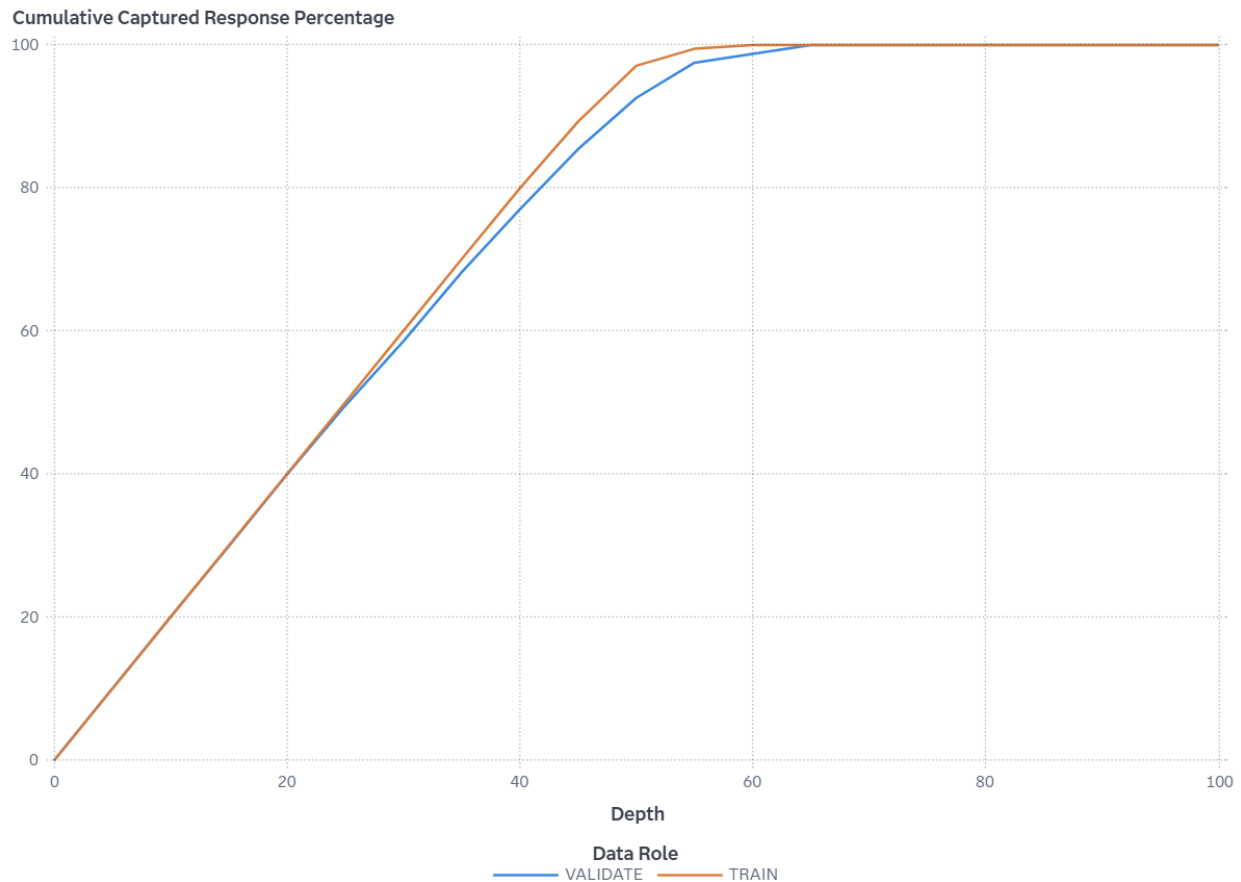


At the 5% quantile (depth of 5), the VALIDATE partition has a Captured response percentage of 10 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 10.04.

At the 5% quantile (depth of 5), the TRAIN partition has a Captured response percentage of 10 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 10.01.

Captured response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event $P_{\text{Attrition_FlagAttrited_Custome}}$, which represents the predicted probability of the event "Attrited Customer" for the target Attrition_Flag . The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Captured response percentage is the percentage of the total number of events that are in that quantile. With no model, it is expected that 5% of the events are in each quantile.

Cumulative Captured Response Percentage



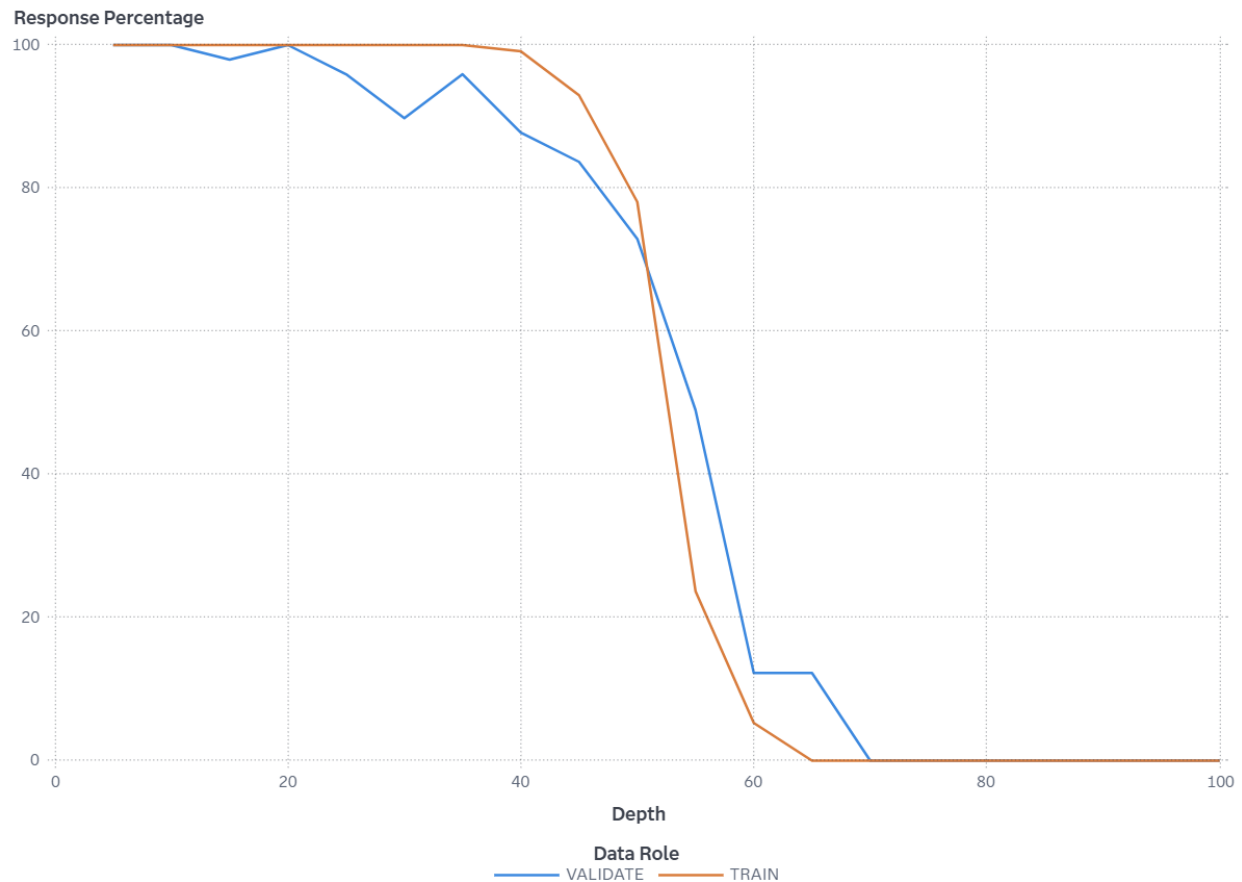
In the top 10% of the data (depth 10), the VALIDATE partition has a Cumulative captured response percentage of 20.1 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 20.08.

In the top 10% of the data (depth 10), the TRAIN partition has a Cumulative captured response percentage of 20 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 20.02.

Cumulative captured response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event `P_Attrition_FlagAttrited_Custome`, which represents the predicted probability of the event "Attrited Customer" for the target `Attrition_Flag`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative captured response percentage for a particular quantile is the percentage of the total number of events that are in the quantiles up to

and including the current quantile. With no model, it is expected that 5% of the events are in each quantile, so the cumulative captured response percentage at depth 10 would be 10%.

Response Percentage

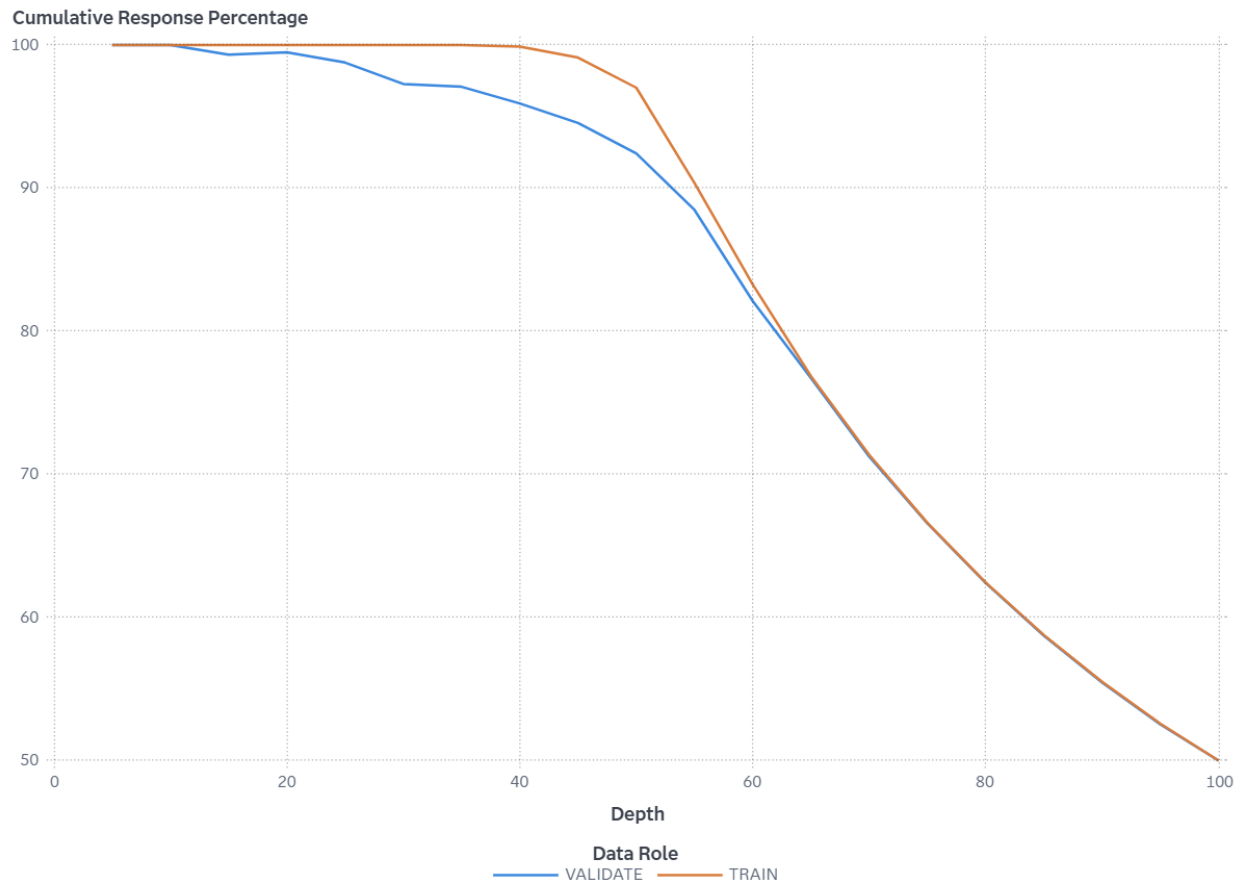


At the 5% quantile (depth of 5), the VALIDATE partition has a Response percentage of 100. The best possible value of Response percentage for this partition at depth 5 is 100.

At the 5% quantile (depth of 5), the TRAIN partition has a Response percentage of 100. The best possible value of Response percentage for this partition at depth 5 is 100.

Response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event $P_{\text{Attrition_FlagAttrited_Custome}}$, which represents the predicted probability of the event "Attrited Customer" for the target Attrition_Flag . The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Response percentage is the percentage of observations that are events in that quantile. With no model, it is expected that the response percentage is constant across quantiles, $100 \times \text{overall-event-rate}$. This is also called the baseline response percentage.

Cumulative Response Percentage



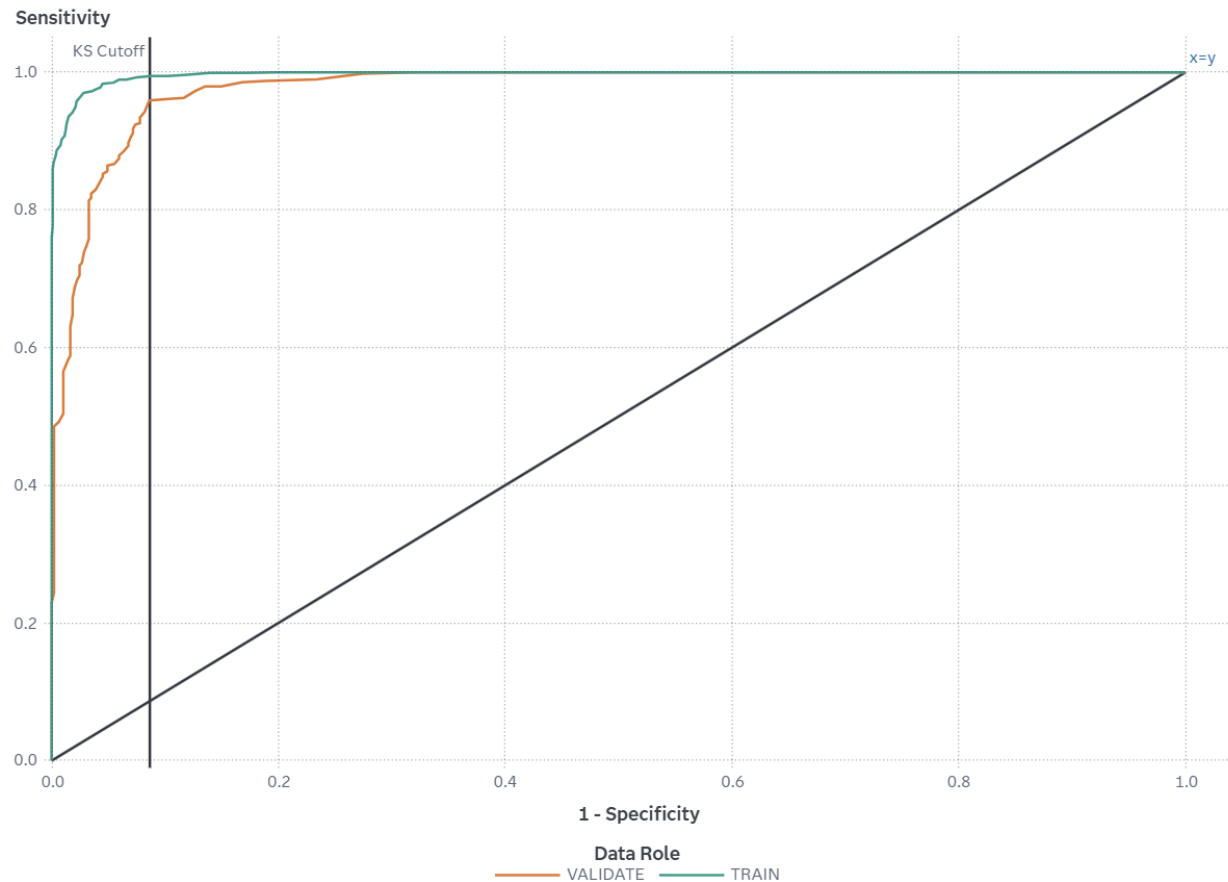
In the top 10% of the data (depth 10), the VALIDATE partition has a Cumulative response percentage of 100. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

In the top 10% of the data (depth 10), the TRAIN partition has a Cumulative response percentage of 100. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

Cumulative response percentage is calculated by sorting in descending order each partition of the data by the predicted probability of the target event `P_Attrition_FlagAttrited_Custome`, which represents the predicted probability of the event "Attrited Customer" for the target `Attrition_Flag`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative response percentage for a particular quantile is the percentage of observations that are events in the quantiles up to and including the current quantile. With no model, it is expected that the response percentage is constant across quantiles, $100 \times \text{overall-event-rate}$. This is also called the baseline response

percentage.

ROC



The ROC curve is a plot of sensitivity (the true positive rate) against 1-specificity (the false positive rate), which are both measures of classification based on the confusion matrix. These measures are calculated at various cutoff values. To help identify the best cutoff to use when scoring your data, the KS Cutoff reference line is drawn at the value of 1-specificity where the greatest difference between sensitivity and 1-specificity is observed for the VALIDATE partition. The KS Cutoff line is drawn at the cutoff value 0.14, where the 1-specificity value is 0.086 and the sensitivity value is 0.959.

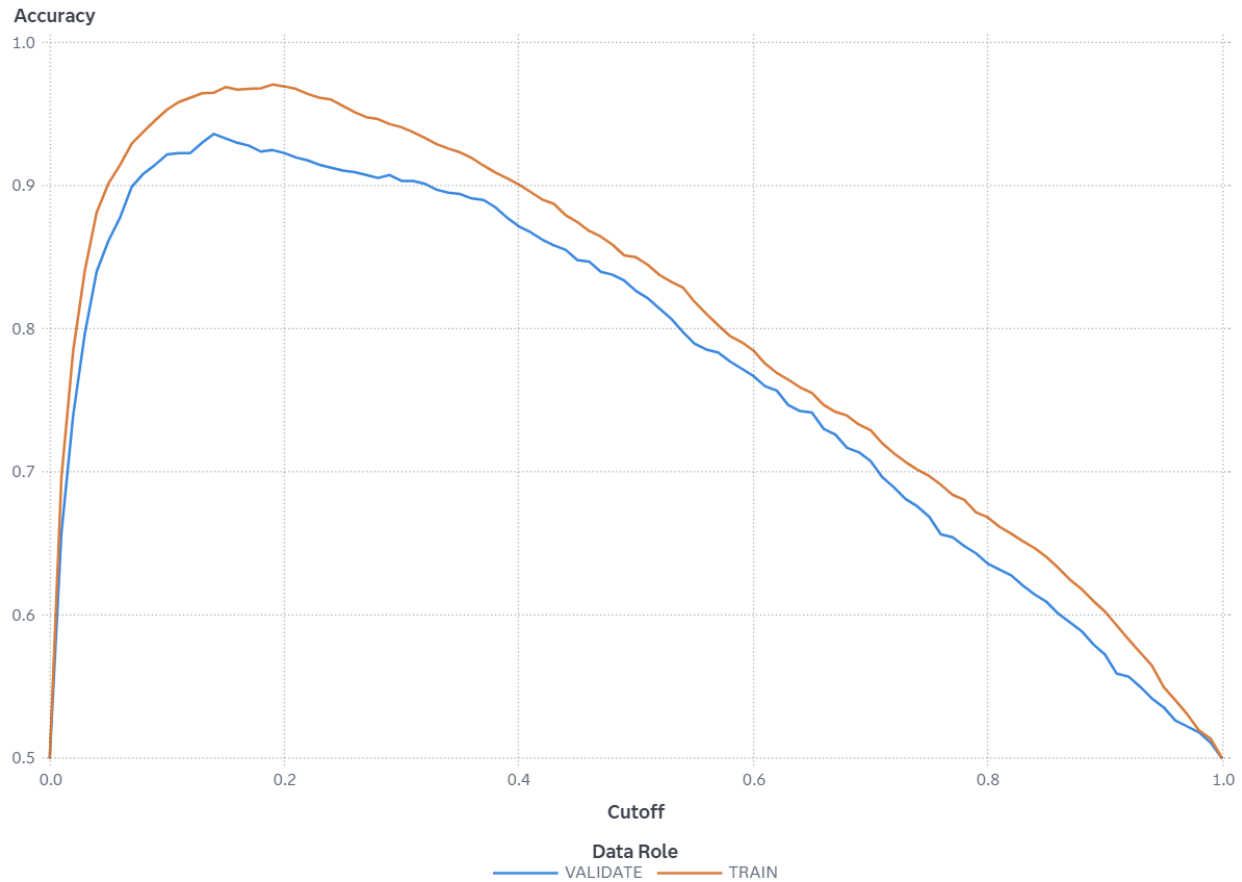
Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether `P_Attrition_FlagAttrited_Custome`, which is the predicted probability of the event "Attrited Customer" for the target `Attrition_Flag`, is greater than or equal to the cutoff value. When `P_Attrition_FlagAttrited_Custome` is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event.

The confusion matrix for each cutoff value contains four cells that display the true positives for events that are correctly classified (TP), false positives for non-events that

are classified as events (FP), false negatives for events that are classified as non-events (FN), and true negatives for non-events that are classified as non-events (TN). True negatives include non-event classifications that specify a different non-event. Sensitivity is calculated as $TP / (TP + FN)$. Specificity, the true negative rate, is calculated as $TN / (TN + FP)$, so 1-specificity is $FP / (TN + FP)$. The values of sensitivity and 1-specificity are plotted at each cutoff value.

A ROC curve that rapidly approaches the upper-left corner of the graph, where the difference between sensitivity and 1-specificity is the greatest, indicates a more accurate model. A diagonal line where sensitivity = 1-specificity indicates a random model.

Accuracy

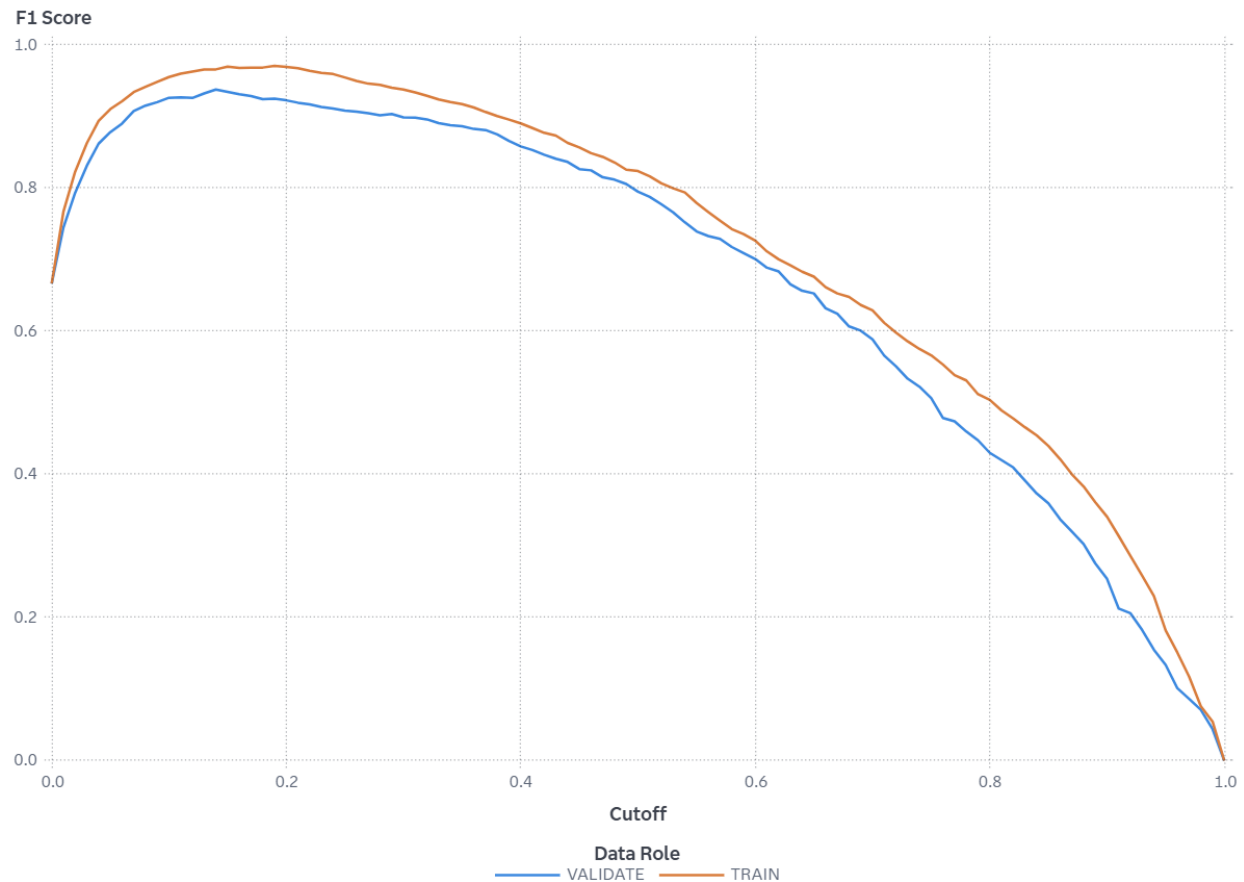


For this model, the accuracy in the TRAIN partition at the cutoff of 0.5 is 0.85.

For this model, the accuracy in the VALIDATE partition at the cutoff of 0.5 is 0.827.

Accuracy is the proportion of observations that are correctly classified as either an event or non-event, calculated at various cutoff values. Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether `P_Attrition_FlagAttrited_Custome`, which is the predicted probability of the event "Attrited Customer" for the target `Attrition_Flag`, is greater than or equal to the cutoff value. When `P_Attrition_FlagAttrited_Custome` is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event. When the predicted classification and the actual classification are both events (true positives) or both non-events (true negatives), the observation is correctly classified. If the predicted classification and actual classification disagree, then the observation is incorrectly classified. Accuracy is calculated as $(\text{true positives} + \text{true negatives}) / (\text{total observations})$.

F1 Score



For this model, the F1 score in the TRAIN partition at the cutoff of 0.5 is 0.824.

For this model, the F1 score in the VALIDATE partition at the cutoff of 0.5 is 0.795.

The F1 score combines the measures of precision and recall (or sensitivity), which are measures of classification based on the confusion matrix that are calculated at various cutoff values. Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether `P_Attrition_FlagAttrited_Custome`, which is the predicted probability of the event "Attrited Customer" for the target `Attrition_Flag`, is greater than or equal to the cutoff value. When `P_Attrition_FlagAttrited_Custome` is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event.

The confusion matrix for each cutoff value contains four cells that display the true positives for events that are correctly classified (TP), false positives for non-events that are classified as events (FP), false negatives for events that are classified as non-events (FN), and true negatives for non-events that are classified as non-events (TN).

True negatives include non-event classifications that specify a different non-event.

Precision is calculated as $TP / (TP + FP)$, and recall (or sensitivity) is calculated as $TP / (TP + FN)$. The F1 score is calculated as $2 * Precision * Recall / (Precision + Recall)$, which is the harmonic mean of Precision and Recall. Larger F1 scores indicate a more accurate model.

Fit Statistics

Target Name	Data Role	Partition Indicator	Formatted Partition
Attrition_Flag	TRAIN	1	1
Attrition_Flag	VALIDATE	0	0

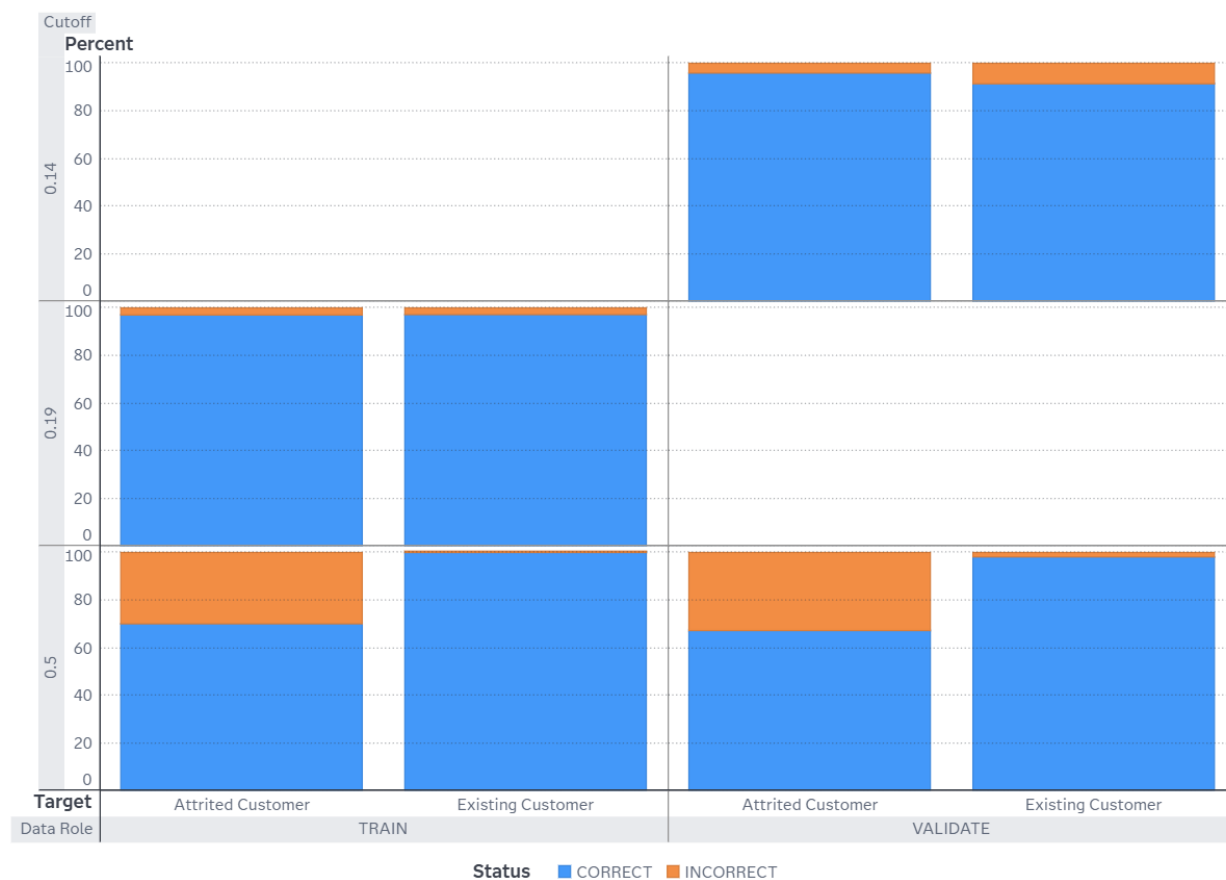
Number of Observations	Average Squared Error	Divisor for ASE	Root Average Squared Error
2,278	0.0962	2,278	0.3102
976	0.1160	976	0.3406

Misclassification Rate	Multi-Class Log Loss	KS (Youden)	Area Under ROC
0.1497	0.2899	0.9421	0.9966
0.1732	0.3495	0.8730	0.9772

Gini Coefficient	Gamma	Tau	KS Cutoff
0.9933	0.9937	0.4968	0.1900
0.9544	0.9560	0.4777	0.1400

KS at User-Specified Cutoff	Misclassification Rate at KS Cutoff (Event)	Misclassification Rate (Event)
0.7006	0.0290	0.1497
0.6537	0.0635	0.1732

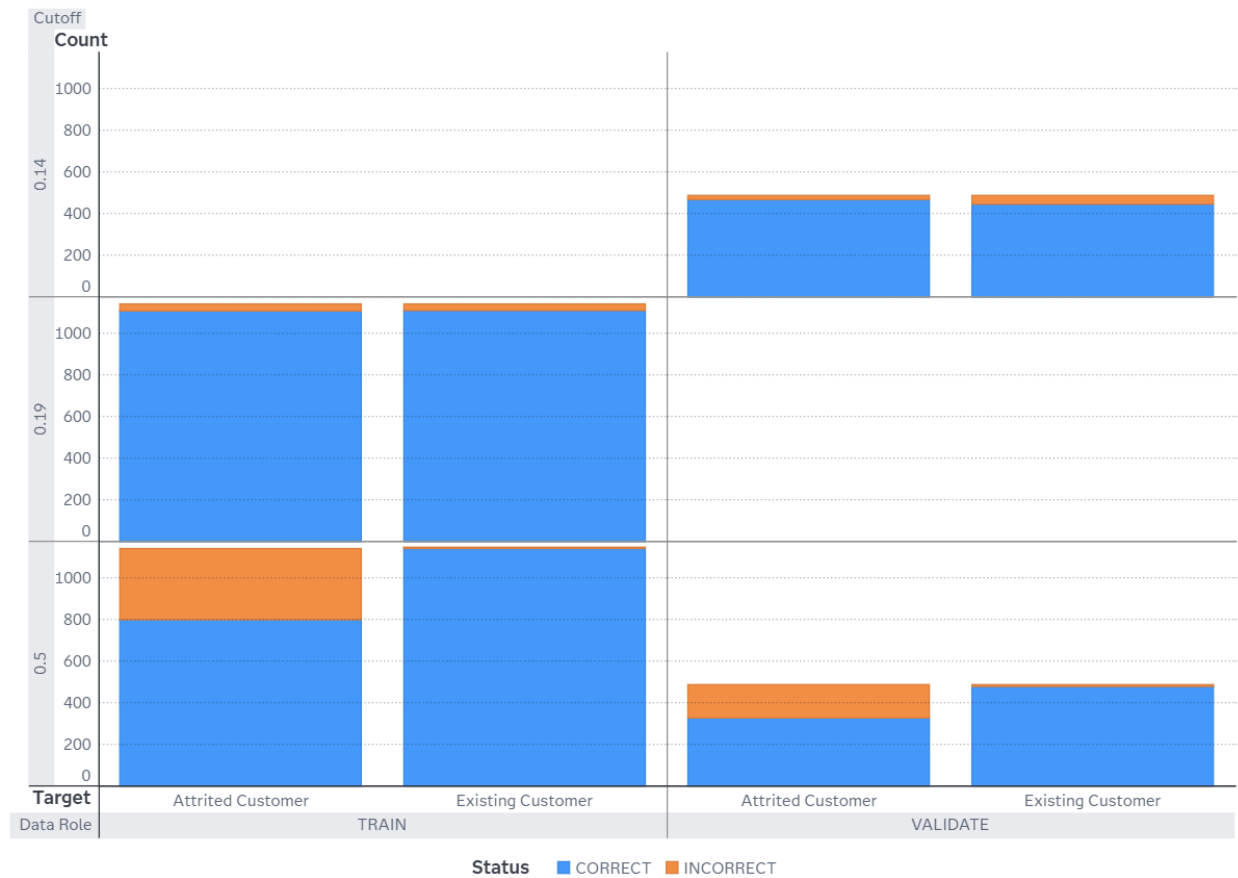
Percentage Plot



The Event Classification report is a visual representation of the confusion matrix at various cutoff values for each partition. The classification cutoffs used in the plot are the default (0.5) and these KS cutoff values for existing partitions: 0.19 (TRAIN), 0.14 (VALIDATE).

For this data, for the bar corresponding to the event level of Attrition_Flag, "Attrited Customer", the segment of the bar colored as "CORRECT" corresponds to true positives.

Count Plot



The Event Classification report is a visual representation of the confusion matrix at various cutoff values for each partition. The classification cutoffs used in the plot are the default (0.5) and these KS cutoff values for existing partitions: 0.19 (TRAIN), 0.14 (VALIDATE).

For this data, for the bar corresponding to the event level of Attrition_Flag, "Attrited Customer", the segment of the bar colored as "CORRECT" corresponds to true positives.

Table

Cutoff	Cutoff Source	Target Name	Response
0.1400	KS	Attrition_Flag	CORRECT
0.1400	KS	Attrition_Flag	INCORRECT
0.1400	KS	Attrition_Flag	CORRECT
0.1400	KS	Attrition_Flag	INCORRECT
0.1900	KS	Attrition_Flag	CORRECT
0.1900	KS	Attrition_Flag	INCORRECT
0.1900	KS	Attrition_Flag	CORRECT
0.1900	KS	Attrition_Flag	INCORRECT
0.5000	Default	Attrition_Flag	CORRECT
0.5000	Default	Attrition_Flag	INCORRECT
0.5000	Default	Attrition_Flag	CORRECT
0.5000	Default	Attrition_Flag	INCORRECT

Event	Value	Training Frequency	Validation Frequency
Attrited Customer	True Positive		468
Attrited Customer	False Negative		20
Existing Customer	True Negative		446
Existing Customer	False Positive		42
Attrited Customer	True Positive	1,105	
Attrited Customer	False Negative	34	
Existing Customer	True Negative	1,107	
Existing Customer	False Positive	32	
Attrited Customer	True Positive	798	328
Attrited Customer	False Negative	341	160
Existing Customer	True Negative	1,139	479
Existing Customer	False Positive	0	9

Test Frequency	Training Percentage	Validation Percentage	Test Percentage
		95.9016	
		4.0984	
		91.3934	
		8.6066	
	97.0149		
	2.9851		
	97.1905		
	2.8095		
	70.0615	67.2131	
	29.9385	32.7869	
	100	98.1557	
	0	1.8443	

Properties

Property Name	Property Value
atAppendLookup	false
atCreateHistory	false
atHistoryLibUri	
atHistoryTblName	
atLeaveAutotuneOn	false
atLookupTableUri	
atMaxBayes	100
atMaxEval	50
atMaxIter	5
atMaxTime	60
atObjectiveInt	ASE
atObjectiveNom	KS
atPopSize	10
atSampleSize	50
atSearchMethod	GA
atTrainProp	0.7000
atUpdateProperties	false
atUseLookup	false
atValidFold	5
atValidMethod	PARTITION
atValidProp	0.3000
atintervalBins	true
atintervalBinsInit	50
atintervalBinsLB	20
atintervalBinsUB	100
atleafSize	false
atleafSizeInit	5
atleafSizeLB	1

Property Name	Property Value
atleafSizeUB	100
atmaxDepth	true
atmaxDepthInit	20
atmaxDepthLB	1
atmaxDepthUB	29
atmaxTrees	true
atmaxTreesInit	100
atmaxTreesLB	20
atmaxTreesUB	150
attrainFraction	true
attrainFractionInit	0.6000
attrainFractionLB	0.1000
attrainFractionUB	0.9000
atvarsToTry	true
atvarsToTryInit	100
atvarsToTryLB	1
atvarsToTryUB	100
autotune_enabled	false
binaryProbCutoff	0.5000
codeLocation	mlearning
criterionMethod	IGR
dataMiningVersion	V2026.03
defaultVarsPerTree	true
exactPctlLift	true
explainFidelity	false
explainInfo	false
fullDatasetReconstitution	false
iCriterionMethod	VARIANCE

Property Name	Property Value
icePlots	false
intBinMethod	QUANTILE
intervalBins	50
leafProp	0.0001
leafSize	5
leafSpec	COUNT
loh	0
maxBranch	2
maxCategories	128
maxDepth	20
maxNumShapVars	20
maxTrees	100
minUseInSearch	1
missingValue	USEINSEARCH
nBins	50
pdNumImportantInputs	5
pdObsSamples	1,000
pdPlots	false
performKernelShap	false
performLime	false
performVI	false
seed	12,345
seedId	12,345
specifyRows	RANDOM
templateRevision	4
train	true
trainFraction	0.6000
truncateLI	5

Property Name	Property Value
truncateUI	95
userProbCutoff	false
varsToTry	100
voteMethod	PROBABILITY

Output

The SAS System

The FOREST Procedure

Model Information	
Number of Trees	100
Number of Variables Per Split	5
Seed	12345
BootstrapPercentage	40
Number of Bits	50
Number of Input Variables	19
MaximumNumber of TreeNodes	257
MinimumNumber of TreeNodes	1
MaximumNumber of Branches	2
MinimumNumber of Branches	2
MaximumDepth	20
MinimumDepth	13
MaximumNumber of Leaves	104
MinimumNumber of Leaves	40
MaximumLeaf Size	314
MinimumLeaf Size	5
ODS MisclassificationRate	0.07330962
Average Number of Leaves	82.5

Training		Validation	Total
Number of Observations Read		2278	976
Number of Observations Used		2278	976

Variable Importance			
Variable	Importance	Std Dev	Relative Importance
Total_Trans_Ct	161.25	78.4780	1.0000
Total_Trans_Amt	123.77	72.8968	0.7676
Total_Revenue_Bd	76.4538	69.6507	0.4761
Total_Ct_Chng_Q4_Q1	34.3428	21.9636	0.2130
Total_Revenue_Mnt	29.2333	13.2752	0.1813
Ass_WebRate_Ratio	27.4702	12.2752	0.1704
Total_Amt_Chng_Q4_Q1	25.9366	12.8737	0.1608
Continents_Count_13_Area	19.6636	4.9314	0.1239
Months_Inactive_12_mon	16.1134	12.7437	0.0999
Customer_Age	12.7238	7.7631	0.0789
Ass_Chng_Ty_Buy	10.7091	7.7634	0.0664
Credit_Limit	8.8243	5.8564	0.0547
Education_Level	8.4021	5.0862	0.0521
Months_in_book	8.1450	5.5882	0.0505
Dependent_Count	6.9021	4.2941	0.0428
Income_Category	5.8444	4.3825	0.0362
Marital_Status	3.2964	2.7318	0.0204
Cash_Category	2.1565	2.7396	0.0134
Gender	1.1841	2.0154	0.0082

Number of Trees	ODS Average Square Error	Training Average Square Error	Validation Average Square Error	Fit Statistics							
				Misclassification Rate	Training Misclassification Rate	Validation Misclassification Rate	Loss	Training Loss	Validation Loss		
1	0.11379	0.0796	0.1095	0.1460	0.0996	0.1460	1.843	1.843	1.843		
2	0.11359	0.0826	0.0889	0.1386	0.0777	0.1035	1.289	0.327	0.761		
3	0.11049	0.0762	0.0820	0.1319	0.0724	0.0984	1.045	0.256	0.644		
4	0.10122	0.0522	0.0799	0.1382	0.0630	0.0948	0.856	0.199	0.403		
5	0.09715	0.0493	0.0766	0.1198	0.0544	0.0953	0.786	0.185	0.354		
6	0.09194	0.0483	0.0735	0.1203	0.0544	0.0984	0.647	0.184	0.334		
7	0.08996	0.0465	0.0745	0.1124	0.0492	0.0942	0.532	0.174	0.333		
8	0.08462	0.0448	0.0714	0.1095	0.0461	0.0973	0.464	0.169	0.302		
9	0.08263	0.0448	0.0714	0.1067	0.0478	0.0922	0.407	0.170	0.303		
10	0.08062	0.0437	0.0704	0.1062	0.0443	0.0922	0.376	0.170	0.302		
11	0.07779	0.0420	0.0690	0.1056	0.0408	0.0902	0.303	0.169	0.280		
12	0.07562	0.0425	0.0681	0.0970	0.0400	0.0892	0.234	0.168	0.257		
13	0.07374	0.0414	0.0674	0.0944	0.0386	0.0902	0.107	0.165	0.255		
14	0.07158	0.0406	0.0662	0.0913	0.0417	0.0892	0.287	0.165	0.253		
15	0.07037	0.0400	0.0661	0.0917	0.0573	0.0890	0.285	0.162	0.254		
16	0.07126	0.0404	0.0671	0.0878	0.0380	0.0890	0.279	0.163	0.257		
17	0.07020	0.0407	0.0673	0.0866	0.0399	0.0889	0.281	0.164	0.259		
18	0.07078	0.0400	0.0664	0.0874	0.0373	0.0889	0.277	0.162	0.256		
19	0.07062	0.0398	0.0668	0.0862	0.0338	0.0779	0.267	0.162	0.251		
20	0.06837	0.0390	0.0649	0.0838	0.0356	0.0789	0.264	0.160	0.252		
21	0.06868	0.0390	0.0666	0.0830	0.0351	0.0779	0.260	0.161	0.252		
22	0.06837	0.0392	0.0661	0.0838	0.0356	0.0738	0.257	0.162	0.251		
23	0.06835	0.0390	0.0637	0.0821	0.0351	0.0738	0.255	0.161	0.249		
24	0.06865	0.0392	0.0626	0.0811	0.0329	0.0738	0.257	0.162	0.249		
25	0.06777	0.0388	0.0630	0.0799	0.0320	0.0748	0.246	0.162	0.238		
26	0.06715	0.0386	0.0629	0.0786	0.0287	0.0748	0.245	0.161	0.237		
27	0.06615	0.0383	0.0624	0.0773	0.0320	0.0748	0.243	0.160	0.236		
28	0.06622	0.0382	0.0620	0.0764	0.0307	0.0737	0.242	0.160	0.225		
29	0.06623	0.0382	0.0621	0.0773	0.0303	0.0737	0.242	0.160	0.225		
30	0.06611	0.0380	0.0619	0.0750	0.0290	0.0707	0.242	0.160	0.225		
31	0.06524	0.0377	0.0618	0.0729	0.0285	0.0738	0.240	0.159	0.224		
32	0.06514	0.0377	0.0619	0.0759	0.0299	0.0748	0.240	0.159	0.225		
33	0.06468	0.0374	0.0616	0.0735	0.0307	0.0738	0.238	0.157	0.224		
34	0.06511	0.0375	0.0618	0.0777	0.0289	0.0738	0.231	0.158	0.225		
35	0.06503	0.0374	0.0618	0.0780	0.0307	0.0738	0.231	0.158	0.225		
36	0.06449	0.0373	0.0617	0.0781	0.0294	0.0737	0.230	0.158	0.225		
37	0.06467	0.0373	0.0616	0.0790	0.0316	0.0737	0.230	0.158	0.224		
38	0.06449	0.0374	0.0616	0.0812	0.0307	0.0737	0.230	0.158	0.225		
39	0.06503	0.0375	0.0615	0.0786	0.0312	0.0737	0.231	0.158	0.224		
40	0.06449	0.0375	0.0616	0.0773	0.0312	0.0737	0.230	0.159	0.224		
41	0.06511	0.0377	0.0618	0.0768	0.0316	0.0737	0.231	0.159	0.225		
42	0.06467	0.0375	0.0616	0.0766	0.0307	0.0737	0.230	0.159	0.224		
43	0.06448	0.0375	0.0616	0.0793	0.0303	0.0707	0.230	0.159	0.224		
44	0.06448	0.0376	0.0612	0.0762	0.0307	0.0686	0.229	0.159	0.223		
45	0.06449	0.0377	0.0614	0.0759	0.0303	0.0687	0.231	0.160	0.223		
46	0.06503	0.0377	0.0621	0.0757	0.0307	0.0707	0.231	0.159	0.223		
47	0.06466	0.0375	0.0613	0.0762	0.0303	0.0687	0.230	0.159	0.223		
48	0.06444	0.0373	0.0608	0.0762	0.0307	0.0707	0.229	0.158	0.222		
49	0.06466	0.0374	0.0609	0.0729	0.0303	0.0737	0.230	0.159	0.223		
50	0.06444	0.0374	0.0607	0.0729	0.0303	0.0707	0.230	0.159	0.222		
51	0.06466	0.0374	0.0610	0.0729	0.0316	0.0737	0.230	0.159	0.223		
52	0.06444	0.0373	0.0609	0.0746	0.0316	0.0707	0.230	0.158	0.223		
53	0.06463	0.0372	0.0608	0.0733	0.0303	0.0687	0.229	0.158	0.222		
54	0.06463	0.0373	0.0610	0.0724	0.0303	0.0707	0.230	0.158	0.223		
55	0.06467	0.0374	0.0611	0.0737	0.0289	0.0707	0.231	0.159	0.223		
56	0.06463	0.0373	0.0610	0.0729	0.0303	0.0737	0.230	0.159	0.223		
57	0.06444	0.0372	0.0610	0.0729	0.0289	0.0687	0.230	0.159	0.223		
58	0.06467	0.0372	0.0611	0.0751	0.0303	0.0707	0.230	0.158	0.223		
59	0.06461	0.0372	0.0611	0.0744	0.0303	0.0687	0.229	0.158	0.223		
60	0.06461	0.0373	0.0613	0.0751	0.0289	0.0687	0.229	0.158	0.224		
61	0.06462	0.0373	0.0613	0.0762	0.0294	0.0737	0.230	0.159	0.224		
62	0.06461	0.0372	0.0612	0.0755	0.0294	0.0737	0.229	0.158	0.224		
63	0.06460	0.0372	0.0611	0.0748	0.0289	0.0737	0.229	0.158	0.223		
64	0.06460	0.0373	0.0612	0.0764	0.0303	0.0687	0.229	0.159	0.224		
65	0.06327	0.0372	0.0610	0.0759	0.0294	0.0687	0.228	0.158	0.223		
66	0.06336	0.0371	0.0609	0.0757	0.0294	0.0707	0.228	0.158	0.222		
67	0.06335	0.0370	0.0610	0.0753	0.0303	0.0737	0.228	0.158	0.222		
68	0.06344	0.0369	0.0609	0.0764	0.0303	0.0707	0.227	0.157	0.221		
69	0.06332	0.0368	0.0609	0.0762	0.0307	0.0707	0.227	0.157	0.222		
70	0.06315	0.0368	0.0608	0.0724	0.0303	0.0707	0.227	0.157	0.222		
71	0.06315	0.0368	0.0608	0.0724	0.0303	0.0687	0.226	0.157	0.221		
72	0.06332	0.0370	0.0608	0.0711	0.0316	0.0687	0.227	0.157	0.221		
73	0.06333	0.0370	0.0609	0.0724	0.0317	0.0686	0.227	0.157	0.222		
74	0.06332	0.0370	0.0608	0.0729	0.0303	0.0687	0.227	0.157	0.222		
75	0.06333	0.0370	0.0609	0.0720	0.0316	0.0687	0.227	0.157	0.222		
76	0.06332	0.0370	0.0609	0.0724	0.0303	0.0687	0.227	0.157	0.222		
77	0.06333	0.0370	0.0609	0.0737	0.0312	0.0687	0.227	0.157	0.222		
78	0.06333	0.0371	0.0610	0.0762	0.0303	0.0687	0.227	0.157	0.222		
79	0.06344	0.0371	0.0610	0.0757	0.0307	0.0707	0.228	0.158	0.222		
80	0.06335	0.0372	0.0611	0.0757	0.0307	0.0687	0.228	0.158	0.223		
81	0.06335	0.0371	0.0610	0.0757	0.0287	0.0707	0.228	0.158	0.223		
82	0.06336	0.0372	0.0612	0.0759	0.0312	0.0707	0.228	0.158	0.223		
83	0.06344	0.0371	0.0611	0.0723	0.0316	0.0687	0.228	0.158	0.223		
84	0.06336	0.0372	0.0610	0.0762	0.0312	0.0687	0.228	0.158	0.223		
85	0.06336	0.0372	0.0613	0.0739	0.0312	0.0686	0.229	0.158	0.223		
86	0.06336	0.0373	0.0614	0.0762	0.0316	0.0687	0.229	0.158	0.224		
87	0.06337	0.0372	0.0614	0.0766	0.0312	0.0687	0.229	0.158	0.224		
88	0.06337	0.0372	0.0614	0.0766	0.0312	0.0687	0.229	0.158	0.224		
89	0.06337	0.0373	0.0614	0.0751	0.0309	0.0679	0.229	0.158	0.224		
90	0.06337	0.0372	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		
91	0.06337	0.0372	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		
92	0.06337	0.0373	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		
93	0.06337	0.0373	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		
94	0.06337	0.0373	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		
95	0.06337	0.0373	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		
96	0.06337	0.0373	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		
97	0.06337	0.0373	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		
98	0.06337	0.0373	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		
99	0.06337	0.0373	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		
100	0.06337	0.0373	0.0614	0.0755	0.0309	0.0679	0.228	0.158	0.224		

